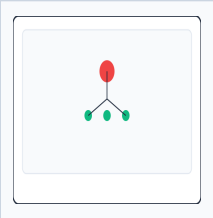
DraftBench — Special Education Resource

Curriculum Aligned & Autism Support Framework



Rutherford's Atomic Model

Subject: Physics | **Target Grade:** 11

Autism-Friendly Learning Framework

*Predictable pacing: One core concept per section.

*Literal, concrete language with real-world grounding.

Step 1: Rutherford's Atomic Model

Core Concept: Discover the nucleus

Explanation: Particles shot at thin gold foil. Most pass through, few bounce back.

Key Takeaways:

• Gold foil experiment

• Positive particles

• Most pass through

Real-World Example: Imagine shooting BBs at a paper sheet.

Communication Support: point to diagram

Vocabulary Builder:

• **nucleus:** center of atom

Step 2: Nucleus and Electrons

Core Concept: Most of atom is empty

Explanation: Protons and neutrons in nucleus. Electrons move around outside.

Key Takeaways:

• Nucleus has protons

• Electrons orbit outside

Real-World Example: Think of sun and planets.

Communication Support: point to diagram

Vocabulary Builder:

• **protons:** positive particles

Step 3: Experiment Results

Core Concept: Few particles bounce back

Explanation: Most particles pass through. Few bounce back, proving nucleus.

Key Takeaways:

• Most particles pass through

• Few particles bounce back

Real-World Example: Imagine shooting BBs at a paper sheet.

Communication Support: point to diagram

Vocabulary Builder:

• **bouncing back:** particles deflect

Step 4: Practice Example

Core Concept: Apply Rutherford's model

Explanation: Draw nucleus and electrons. Label protons and neutrons.

Key Takeaways:

• Draw nucleus and electrons

• Label protons and neutrons

Real-World Example: Draw nucleus and electrons on paper.

Communication Support: point to diagram

Vocabulary Builder:

• **neutrons:** neutral particles

Step 5: Review Rutherford's Model

Core Concept: Nucleus and electrons

Explanation: Nucleus has protons. Electrons orbit outside.

Key Takeaways:

• Nucleus has protons

• Electrons orbit outside

Real-World Example: Think of sun and planets.

Communication Support: point to diagram

Vocabulary Builder:

• **orbit:** path around nucleus

Curriculum Reference Standards

• Physics Grade 11 एलेक्ट्रकि भाइल्ड र उनको फरक (1. Reduced- Grde-11-Physics_ogc1tkt.pdf)

• Physics Grade 11 इलेक्ट्रोमोटवि शक्ति (emf) र वभिव अंतर (p.d.) (1. Reduced- Grde-11-Physics_ogc1tkt.pdf)

• Physics Grade 11 यादरूपी अभ्यास (1. Reduced- Grde-11-Physics_ogc1tkt.pdf)